

AI Engineering Roadmap

9 phases · 16+ DeepLearning.ai courses

Overview

This roadmap takes a practicing full-stack developer through every layer of modern AI engineering — from Python fundamentals to production deployment. Each phase builds directly on the previous one. Courses are sourced from DeepLearning.ai (free to audit) and are ordered by dependency, not difficulty.

At a glance

Phase	Topic	Duration
1	Python fundamentals Core syntax & dev environment	~1-2 weeks
2	Data extraction & knowledge engineering Gather, clean & structure data for LLMs	~1-2 weeks
3	LangChain & LangGraph Chains, memory, agents, state graphs	~1 week
4	AI tools — RAG & fine-tuning Vector search, retrieval pipelines, model adaptation	~3 weeks
5	Prompt engineering System prompts, CoT, structured outputs	~3 weeks
6	AI agents & decision systems ReAct, tool use, planning, multi-agent	~2 weeks
7	Deployment — vLLM & NanoLLM Serve, quantize & scale LLMs in production	~2 weeks
8	Evaluation, MLOps & LLMOps Observe, evaluate & iterate AI systems	~2 weeks

How to use this plan

1. Spend the most time on Phases 4-5-6. RAG, fine-tuning, and agents are where AI engineering diverges from regular software work.
2. One DeepLearning.ai course per week (1-4 hours each), then build a mini-project applying what you learned.
3. All courses are free to audit at deeplearning.ai

Detailed Phase Breakdown

1

Python fundamentals

Core syntax & dev environment

~1-2 weeks

DEEPLEARNING.AI COURSES

Course	What you'll learn
AI Python for Beginners	Andrew Ng — perfect entry point even for devs
** Even as a full-stack dev, run through this quickly — Python idioms differ from JS/TS significantly.	

2

Data extraction & knowledge engineering

Gather, clean & structure data for LLMs

~1-2 weeks

DEEPLEARNING.AI COURSES

Course	What you'll learn
Preprocessing Unstructured Data for LLM Applications	PDFs, images, tables — convert to LLM-ready format
Knowledge Graphs for RAG	Structuring domain knowledge for retrieval pipelines
** Data quality is 80% of RAG quality. Invest heavily here — it multiplies the value of every later phase.	

3

LangChain & LangGraph

Chains, memory, agents, state graphs

~2 weeks

DEEPLEARNING.AI COURSES

Course	What you'll learn
LangChain for LLM Application Development	Core chains, agents, memory — the foundation course
Functions, Tools and Agents with LangChain	Tool calling, OpenAI functions, agent patterns
** Master LCEL first — it unifies chains, agents, and tools under one composable interface.	

4

AI tools — RAG & fine-tuning

Vector search, retrieval pipelines, model adaptation

~3 weeks

DEEPLARNING.AI COURSES

Course	What you'll learn
Building and Evaluating Advanced RAG	Sentence window, auto-merging, evaluation pipelines
Vector Databases: from Embeddings to Applications	Weaviate-based deep dive on similarity search
Finetuning Large Language Models	When and how to fine-tune vs prompt-tune vs RAG
** Start with RAG before fine-tuning — it solves 90% of use-cases at far lower cost and complexity.	

5

Prompt engineering

System prompts, CoT, structured outputs

~1 week

DEEPLARNING.AI COURSES

Course	What you'll learn
ChatGPT Prompt Engineering for Developers	OpenAI x DeepLearning.ai — the foundational reference
Prompt Engineering with Llama 2 & 3	Open-source model prompting techniques
** Apply prompting patterns directly to your real tasks — don't study these in isolation from actual problems.	

6

AI agents & decision systems

ReAct, tool use, planning, multi-agent

~3 weeks

DEEPLARNING.AI COURSES

Course	What you'll learn
AI Agents in LangGraph	Production-grade agent patterns with state machines
Multi AI Agent Systems with crewAI	Role-based multi-agent collaboration frameworks
Building Agentic RAG with LlamaIndex	Query planning agents and recursive retrieval
Agentic AI	Integrate AI with external tools
** Think of agents as state machines — LangGraph makes this intuitive and debuggable for engineers.	

7

Deployment — vLLM & NanoLLM

Serve, quantize & scale LLMs in production

~2 weeks

DEEPLARNING.AI COURSES (Advanced)

Course	What you'll learn
Efficiently Serving LLMs	KV cache, continuous batching, throughput optimization
Quantization Fundamentals with Hugging Face	GGUF, AWQ, GPTQ — reduce model size without quality loss
** vLLM is the industry standard for GPU inference servers. NanoLLM targets Jetson / edge deployments.	

8

Evaluation, MLOps & LLMOps

Observe, evaluate & iterate AI systems

~2 weeks

DEEPLARNING.AI COURSES (Advanced)

Course	What you'll learn
LLMOps	CI/CD pipelines, versioning, monitoring for LLM applications
Evaluating and Debugging Generative AI	W&B tracing, evaluation datasets, metrics
Automated Testing for LLMOps	Test pipelines for non-deterministic AI outputs
** Ship fast, but measure everything. LLM outputs are probabilistic — evaluation is non-negotiable in production.	

Priority Courses

If you can only complete three courses, these deliver the highest return:

#	Course	Why it matters
1	AI Agents in LangGraph	The most practical course for production agent systems
2	Building and Evaluating Advanced RAG	Most widely applicable AI engineering skill
3	Efficiently Serving LLMs	Separates engineers who deploy from those who only prototype

deeplearning.ai

All courses listed are free to audit. Generated for Ibrahim — AI YouTube Content Creator.